

Value Change Magnitude Module (VCMM)

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Canonical Definition

Value Change Magnitude Module (VCMM) is the third Core Module of the Value Change Standard (VCS) within VFM® — Value Flow Mechanisms Architecture.

VCMM governs the determination of the extent, scale, intensity, proportion, significance, or measurable degree of an identified and classified Value Change within its defined Value Context.

The Module establishes a controlled measurement layer through which Value Change may be expressed quantitatively, qualitatively, proportionally, comparatively, categorically, probabilistically, or through another governed measurement method appropriate to the value being assessed.

VCMM does not require all value to be expressed in monetary terms and does not assume that one measurement system can legitimately represent every form of value.

Its foundational task is:

to determine how much value has materially changed, using a measurement basis appropriate to the governed value and without confusing magnitude with cause, price, or resulting Value State.

Architectural Position

VCMM is the third module within VCS.

It follows:

VCTM — Value Change Type Module

and precedes:

VCCM — Value Change Causation Link Module.

The internal sequence is:

Governed Value State → VCIM — Value Change Identification → VCTM — Value Change Type → VCMM — Value Change Magnitude

→ VCCM — Value Change Causation Link → VCRM — Value Change Record

VCIM establishes:

that a material Value Change occurred.

VCTM establishes:

what kind of Value Change occurred.

VCMM establishes:

how large or significant that Value Change is.

Internal Module Flow

Identified Value Change → Governed Value Change Type → Affected Value Dimension → Reference Basis Selection → Measurement Method Selection → Baseline Establishment → Change Measurement → Uncertainty Assessment → Materiality Assessment → Magnitude Determination → Governed Value Change Magnitude → VCCM — Value Change Causation Link Module

Operational Space

VCMM governs the measurement space between:

classified Value Change

and

causal linkage of that change.

Its operational space includes:

absolute magnitude; relative magnitude; proportional magnitude; qualitative magnitude; quantitative magnitude; monetary magnitude; non-monetary magnitude; multidimensional magnitude; cumulative magnitude; incremental magnitude; comparative magnitude; threshold-based magnitude; range-based magnitude; uncertainty-bounded magnitude; materiality; measurement confidence; baseline selection; reference-period selection; normalization; magnitude aggregation; magnitude decomposition.

Magnitude Requires a Reference Basis

A statement that value changed by:

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has little governed meaning unless the reference basis is known.

VCMM therefore requires magnitude to remain connected to an appropriate reference basis.

The reference basis may include:

previous Value State; defined baseline; initial state; expected state; target state; benchmark; comparable object; previous period; scenario baseline; operational requirement; resource baseline; replacement basis; governed index; another explicitly defined reference point.

Therefore:

Magnitude without Reference Basis $\neq$ Governed Value Change Magnitude

Absolute Value Change Magnitude

VCMM may express change as an absolute difference.

For a measurable value dimension:

$$\Delta V = V_2 - V_1$$

where:

V_1 = governed reference value

and

V_2 = governed subsequent value.

Examples:

Production Capacity: 1,000 units/day $\rightarrow$ 1,250 units/day

Absolute Change: +250 units/day

or:

Energy Consumption: 10 MWh $\rightarrow$ 8 MWh for equivalent governed output

Absolute Change: -2 MWh.

The meaning of the change remains dependent upon the governed Value Context.

Relative Value Change Magnitude

Where appropriate, VCMM may determine relative change:

$$\text{Relative Change} = (V_2 - V_1) / V_1$$

This permits expressions such as:

+25% production capacity

-20% energy consumption

+12% availability

-8% operational capability.

Percentage movement is useful only where the underlying measurement basis legitimately supports proportional comparison.

Magnitude $\neq$ Direction

VCTM determines whether a change is:

positive, negative, neutral structural, mixed, or indeterminate.

VCMM determines its magnitude.

Therefore:

Direction $\neq$ Magnitude

A negative change can be small or catastrophic.

A positive change can be marginal or transformative.

VCMM must preserve this distinction.

Magnitude ≠ Materiality

A measurable Value Change is not necessarily materially significant.

For example:

0.01% change

may be immaterial in one context.

The same change may be critical in another if it affects:

spacecraft navigation; high-frequency financial infrastructure; structural safety; medical manufacturing; energy-grid stability; mission-critical computation.

VCMM therefore distinguishes:

Measured Magnitude

from

Governed Materiality.

Materiality must be assessed relative to the purpose, beneficiary, object, operational context, and applicable thresholds.

Qualitative Magnitude

Not every Value Change can be legitimately expressed numerically.

VCMM therefore supports qualitative magnitude states.

A governance framework may use states such as:

Negligible

Minor

Moderate

Major

Severe

Critical

Transformative

or another domain-specific classification.

The categories must have defined criteria.

Qualitative measurement must not become arbitrary merely because numerical measurement is unavailable.

Quantitative Magnitude

Where reliable measurement is possible, VCMM supports quantitative magnitude.

Possible measurement forms include:

currency; percentage; physical units; energy units; production units; time; computational resources; capacity; throughput; availability; reliability; performance; efficiency; probability; information quantity; resource consumption; another governed metric.

VCMM does not privilege monetary measurement over other legitimate measurement forms.

Monetary Magnitude

Where Value Change can legitimately be represented economically, VCMM may express magnitude in monetary terms.

Examples include:

CHF +2.4 million

EUR -600,000

USD +15 million

or another currency.

However:

Monetary Magnitude $\neq$ Total Value Change

where the governed value contains material non-monetary dimensions.

A monetary measure is one representation of magnitude, not a universal definition of value.

Non-Monetary Magnitude

VCMM explicitly supports non-monetary Value Change.

Examples include:

Energy Requirement: -18%

Computational Requirement: -30%

Production Capacity: +22%

System Availability: +4.5 percentage points

Mission Range: +15%

Data Utility: materially reduced

Operational Resilience: major improvement

Resource Accessibility: severe reduction.

This capability is fundamental to VFM® because value frequently changes before that change is monetized.

Multi-Dimensional Magnitude

A Value Change may affect several value dimensions simultaneously.

VCMM must therefore support:

Multi-Dimensional Magnitude Determination.

For example:

AI Infrastructure Optimization

may produce:

Compute Consumption: -25%

Energy Consumption: -18%

Latency: -12%

Operational Capacity: +30%

Infrastructure Cost: -15%

Provider Dependency: +20%.

VCMM should not automatically collapse these dimensions into one synthetic score.

Each magnitude may remain separately governed until a legitimate aggregation methodology exists.

Magnitude Aggregation

Where multiple Value Changes are aggregated, VCMM requires the aggregation basis to be explicit.

Aggregation may require:

common units; normalization; weighting; conversion; defined equivalence; temporal alignment; beneficiary alignment; contextual compatibility.

Therefore:

Multiple Magnitudes $\neq$ Automatically Additive Magnitudes

CHF 1 million of economic value cannot simply be added to a 20% resilience improvement or 15% energy reduction without a governed conversion basis.

Magnitude Decomposition

A large Value Change may consist of several smaller changes.

VCMM may decompose:

Total Value Change Magnitude

into:

Component A

Component B

Component C

where the measurement structure permits it.

This is especially useful for:

portfolios; industrial systems; infrastructure; organizations; complex digital systems; AI systems; supply chains; banking assets; mission systems.

Decomposition preserves visibility into where the magnitude originates before VCMM examines causation.

Incremental Value Change

VCMM may determine the magnitude of an incremental change.

Example:

Efficiency

Period 1 → +3%

Period 2 → +2%

Period 3 → +4%.

Each change can be separately governed.

This prevents cumulative outcomes from obscuring individual Value Change events.

Cumulative Value Change Magnitude

Where repeated changes accumulate, VCMM may determine cumulative magnitude across a defined period.

The cumulative basis must identify:

starting reference state; relevant changes; time interval; compounding method; aggregation method; excluded changes; measurement uncertainty.

Cumulative magnitude must not be inferred by simply adding percentages where compounding or changing baselines materially affect the result.

Comparative Magnitude

VCMM may compare Value Change across:

objects; assets; systems; periods; scenarios; organizations; technologies; portfolios; operational configurations.

Example:

System A: Energy reduction –12%

System B: Energy reduction –28%.

Comparability requires sufficiently compatible measurement bases.

Where bases differ materially, VCMM should identify the comparison as limited or non-equivalent.

Normalized Magnitude

Normalization may be required where objects differ in scale.

Examples include:

Energy per unit produced

Cost per transaction

Compute per inference

Output per employee-hour

Revenue per deployed asset

Failure rate per operating hour.

Normalization can reveal Value Change that absolute measurement alone would conceal.

VCMM requires the normalization basis to remain explicit.

Baseline Governance

Baseline selection can materially alter the apparent magnitude of Value Change.

VCMM therefore requires the baseline to be:

identifiable; relevant; temporally defined; methodologically appropriate; reproducible where applicable; resistant to opportunistic selection; documented.

Changing the baseline after observing the result can artificially enlarge or reduce apparent Value Change.

Therefore:

Baseline Manipulation = Magnitude Governance Risk

Measurement Method

The selected measurement method must correspond to the value dimension being assessed.

Possible methods include:

direct measurement; calculated measurement; comparative measurement; model-based estimation; probabilistic estimation; expert assessment; index-based measurement; proxy measurement; composite measurement; hybrid measurement.

The method must remain visible in the magnitude determination.

Measured ≠ Estimated

VCMM distinguishes between:

Measured Magnitude

and

Estimated Magnitude.

A direct operational measurement and a model-derived estimate may both be useful.

They are not methodologically identical.

Estimated magnitude should identify the estimation basis and material uncertainty.

Magnitude Range

Where a single-point estimate would create false precision, VCMM may represent magnitude as a range.

Example:

Estimated Economic Value Change: CHF +8.0M to +11.5M

or:

Expected Efficiency Improvement: +14% to +19%.

Range-based representation is preferable where uncertainty is material.

Probabilistic Magnitude

Where appropriate, magnitude may be represented probabilistically.

For example:

70% probability that the Value Change lies between +12% and +18%.

Probabilistic representation must identify the underlying method and assumptions.

Probability does not eliminate uncertainty.

It structures it.

Uncertainty-Bounded Magnitude

VCMM requires material uncertainty to remain visible.

Uncertainty may arise from:

incomplete measurement; uncertain baseline; data limitations; estimation method; model dependence; volatility; temporal instability; measurement error; conversion assumptions; aggregation assumptions.

A governed magnitude may therefore take the form:

Magnitude Estimate + Uncertainty Range + Measurement Basis.

Magnitude Confidence

Organizations may assign confidence to a magnitude determination.

Possible states may include:

Low Confidence

Moderate Confidence

High Confidence

or another governed scale.

Confidence must not substitute for the underlying magnitude.

It expresses the strength of the measurement basis.

Threshold-Based Magnitude

VCMM may define thresholds for operational classification.

Example:

< 2% → Minor

2–10% → Moderate

10–25% → Major

> 25% → Severe

These thresholds are illustrative only.

Each implementation must establish thresholds appropriate to the governed context.

VFM® does not impose one universal materiality scale across all forms of value.

Magnitude and Time

The same magnitude may have different meaning depending on how quickly it occurs.

For example:

–20% over ten years

and

–20% in one hour

represent materially different operational realities.

VCTM may classify temporal character.

VCMM may preserve the relevant time basis when measuring magnitude.

Therefore, where relevant:

Magnitude + Time Basis

should remain connected.

Rate of Value Change

Where operationally useful, VCMM may determine the rate at which value changes.

Examples:

CHF –500,000/month

Capacity +4%/quarter

Energy Efficiency +1.2%/year

Asset Utility –3%/1,000 operating hours.

Rate measurement supports monitoring and later lifecycle governance.

Magnitude Threshold Crossing

A Value Change may become operationally significant when it crosses a defined threshold.

VCMM may identify:

Threshold Not Reached

Threshold Approaching

Threshold Reached

Threshold Exceeded.

The threshold itself may originate from organizational, contractual, technical, regulatory, financial, or other governance systems.

VCMM governs its relationship to Value Change magnitude, not the external authority that created it.

Magnitude Reassessment

Magnitude may require reassessment when:

new measurements become available; baseline data changes legitimately; an estimate becomes directly measurable; uncertainty narrows; additional Value Change occurs; measurement error is discovered; the governed context materially changes.

A magnitude determination is therefore versionable.

VCMM Boundary with VCTM

VCTM determines:

what kind of Value Change occurred.

VCMM determines:

how much it changed.

Therefore:

Value Change Type $\neq$ Value Change Magnitude

VCMM Boundary with VCCM

VCMM determines magnitude.

VCCM establishes the governed relationship between Value Change and relevant causes.

Therefore:

Magnitude $\neq$ Causation

A CHF 10 million Value Change does not itself establish why the change occurred.

VCMM Boundary with VSS

VSS determines the current governed Value State.

VCMM determines the magnitude of movement between relevant value conditions.

Therefore:

Resulting State $\neq$ Magnitude of Change

Two objects may reach the same Value State after experiencing materially different Value Change magnitudes.

VCMM Boundary with External Valuation

Systems

VCMM may consume or produce information relevant to:

financial valuation; accounting; banking; insurance; taxation; investment; blockchain; industrial measurement; AI infrastructure; energy systems; scientific systems.

It does not replace domain-specific valuation, accounting, tax, legal, or regulatory methodologies.

VCMM governs the magnitude of Value Change within VFM®.

Minimum Implementation Framework

1. Value Change Input

Receive the identified and classified Value Change from VCIM and VCTM.

2. Value Dimension

Identify the exact value dimension whose magnitude is being determined.

3. Reference Basis

Establish the baseline, previous state, benchmark, target, or other governed reference basis.

4. Measurement Method

Select the appropriate quantitative, qualitative, comparative, probabilistic, monetary, non-monetary, or hybrid measurement method.

5. Measurement Unit

Define the unit, scale, classification, index, or other representation used.

6. Magnitude Determination

Determine the absolute, relative, proportional, categorical, range-based, or other appropriate magnitude.

7. Uncertainty Assessment

Identify material uncertainty affecting the determination.

8. Materiality Assessment

Determine whether the magnitude is materially significant within the governed Value Context.

9. Confidence / Qualification

Where appropriate, establish confidence or qualification attached to the magnitude.

10. Magnitude Record

Create a governed record suitable for downstream causation linkage and Value Change recording.

Governance Outputs

VCMM may produce:

Value Change Magnitude Record Value Change Baseline Record Value Change Measurement Basis Value Change Measurement Method Absolute Value Change Magnitude Relative Value Change Magnitude Qualitative Value Change Magnitude Monetary Value Change Magnitude Non-Monetary Value Change Magnitude Multi-Dimensional Magnitude Record Cumulative Value Change Magnitude Normalized Value Change Magnitude Value Change Materiality State Value Change Uncertainty Range Value Change Magnitude Confidence Governed Value Change Magnitude

These outputs provide the governed input for:

VCCM — Value Change Causation Link Module.

Use Case 1 — Banking: Detecting Material

Collateral Value Deterioration Before Price Alone

Shows It

A bank finances a large industrial production asset.

The asset remains operational and its most recent external valuation has not yet materially changed.

Operational data nevertheless shows:

Production Capacity: -14%

Energy Efficiency: -11%

Unplanned Downtime: +27%

Maintenance Requirement: +32%.

VCTM has already classified several negative operational and efficiency Value Changes.

VCMM now determines their magnitude against the governed baseline.

Instead of waiting for the changes to appear as one delayed market-price or accounting adjustment, the bank can preserve the separate magnitude of each Value Change.

This creates a governed basis for:

collateral monitoring, credit-risk review, covenant assessment, refinancing analysis, impairment investigation, and early-warning systems.

The key capability is:

material Value Change can become measurable before it becomes fully visible in market price.

Use Case 2 — AI Infrastructure: Measuring Value

Created Through Cost Reduction

An organization operates an AI platform requiring:

100 million computational units per month

and

1,000 MWh of energy

to deliver a defined operational workload.

After optimization, equivalent governed output requires:

72 million computational units

and

810 MWh of energy.

VCMM determines:

Computational Requirement Change: -28%

Energy Requirement Change: -19%.

If the organization also establishes an appropriate economic conversion basis, VCMM may separately determine:

Infrastructure Cost Change: -CHF X

and

Economic Value Change: +CHF Y.

The non-monetary and monetary magnitudes remain separately visible.

The organization can therefore determine not merely that optimization created value, but:

how much resource, operational, and economic value changed.

This supports:

AI cost governance, infrastructure planning, investment decisions, capacity allocation, ROI analysis, energy governance, and technology comparison.

Enterprise Application

VCMM provides the measurement layer required to move VFM® from:

“value changed”

to:

“value changed by this governed magnitude, against this reference basis, using this method, with these limitations and this uncertainty.”

This makes the Module directly applicable to:

banking; investment; insurance; industrial production; infrastructure; AI systems; computational infrastructure; energy; digital assets; blockchain; intellectual property; organizational capability; supply chains; public assets; mission systems; scientific systems; future value-based economic models.

VCMM is particularly valuable where organizations need to govern forms of Value Change that are real and operationally significant but not yet fully monetized.

Foundational Principles

Value Change magnitude requires a governed reference basis.

Magnitude is distinct from direction.

Magnitude is distinct from materiality.

Magnitude is distinct from causation.

Magnitude is distinct from resulting Value State.

Value Change does not need to be monetary to be measurable.

Monetary magnitude is not a universal representation of total value.

Different value dimensions should remain separately measurable where aggregation would destroy meaning.

Multiple magnitudes must not be aggregated without a governed equivalence or conversion basis.

Measured and estimated magnitudes must remain distinguishable.

Uncertainty must remain visible where measurement is uncertain.

A precise number is not automatically an accurate Value Change magnitude.

The measurement method must fit the value dimension being measured.

A Value Change becomes analytically useful when its scale can be understood without losing the context that gives that scale meaning.

Canonical Closing Statement

Value Change Magnitude Module establishes the measurement layer of the Value Change Standard by determining the extent, scale, intensity, proportion, significance, or measurable degree of an identified and classified Value Change against an explicit governed reference basis. Through governance of baselines, measurement methods, units, absolute and relative change, qualitative and quantitative magnitude, monetary and non-monetary magnitude, multidimensional measurement, normalization, aggregation, decomposition, cumulative change, materiality, uncertainty, confidence, thresholds, temporal basis, and reassessment, VCMM enables VFM® to represent how much value has changed without reducing all value to market price, currency, accounting treatment, or a single universal metric. By separating magnitude from direction, causation, materiality, and resulting Value State, VCMM creates a reusable measurement layer through which operational, economic, physical, digital, strategic, computational, infrastructural, and other forms of Value Change can be compared, monitored, analyzed, and transferred into downstream governance. A Value Change

should not be treated as fully characterized merely because its direction is known; its governed magnitude must establish how much changed, relative to what, by which method, and with what uncertainty.

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Governed Links

→ Value Change Standard (VCS)

→ VFM® Architecture Map

→ Previous: Value Change Type Module (VCTM)

→ Next: Value Change Causation Link Module (VCCM)

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Canonical Source

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